

METHODS OF PROOF - HOMEWORK

1. Give a direct proof that
 $(n \text{ and } m \text{ are even integers}) \Rightarrow (m + n \text{ is an even integer})$
is true.
2. Give a proof by contradiction to show that
 $n + m \text{ is an odd integer} \Rightarrow \text{precisely one of } n \text{ and } m \text{ is an odd integer}$
is true.
3. Prove that
 $n \text{ is odd and } m \text{ is even} \Rightarrow 3n^2 + 5m^2 - 7mn + 9 \text{ is even}$
is true.
4. Prove that if x is an even integer and y is a multiple of 3 then $5xy - 6x + 2y^2 - 18$ is a multiple of 6.
5. Prove that if c is divisible by b and b is divisible by a then c is divisible by a .
6. Prove by contradiction that if $a^2 - 2a + 7$ is odd, then a is even.
7. Prove by contradiction that there does not exist a smallest positive rational number.
8. Prove by contradiction that
 - (a) If x^2 is multiple of 3 then x is also a multiple of 3.
 - (b) $\sqrt{3}$ is irrational.
9. Prove by contradiction that
 - (a) If x^3 is even then x is even.
 - (b) $\sqrt[3]{2}$ is irrational.

10. Prove each of the following statements by induction.

(a) $\sum_{i=1}^n (2i - 1) = n^2$ for all $n \geq 1$.

(b) $\sum_{i=1}^n (4i - 3) = n(2n - 1)$ for all $n \geq 1$.

(c) $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$.

(d) $\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$ for all $n \geq 1$.

(e) $\sum_{i=1}^n 2^i = 2^{n+1} - 2$ for all $n \geq 1$.

(f) $\sum_{i=1}^n \frac{1}{(2i-1)(2i+1)} = \frac{n}{2n+1}$ for all $n \geq 1$.

(g) $\sum_{i=1}^n i(i!) = (n+1)! - 1$ for all $n \geq 1$.

(h) $4^n + 5$ is divisible by 3 for all $n \geq 1$.

(i) $2^{2n} - 1$ is divisible by 3 for all $n \geq 1$.

(j) $9^n - 2^n$ is divisible by 7 for all $n \geq 1$.

(k) $(n^3 - n)$ is divisible by 3 for all $n \geq 1$.